

## **Experiment 09: Write an application that draws basic graphical primitives in the screen**

### **PART A**

**A.1 Aim:** To write an application that draws basic graphical primitives in the screen.

**A.2 Objectives:** To introduce students with various tools like Android Studio, NS2, Wire-shark, Cisco packet tracer, WAP supported browser etc.

**A.3 Outcome:** After successful completion of this experiment students will be able to develop an application that draws basic graphical primitives in the screen.

### **A.4 Theory:**

#### **SOFTWARE:**

- Android Studio
- The Android SDK (Starter Package)
- Gradle
- Java Development Kit (JDK) 5

#### **DESCRIPTION:**

8. Open android studio and select new android project .
9. Give project name and select next
10. Choose the android version.
11. Enter the package name. package name must be two word separated by comma and click finish
12. Go to package explorer in the left hand side and select our project.

13. Go to res folder and select layout. Double click the main.xml file 14. Now you can see the Graphics layout window.

### SourceCode:

#### CodeforActivity\_main.xml:

```
<?xmlversion="1.0"encoding="utf-8"?>

<RelativeLayoutxmlns:android="http://schemas.android.com/apk
/res/android"android:layout_width="match_parent"android:layo
ut_height=" match_parent">

<ImageView
    android:layout_width="match_pare
    nt"android:layout_height="match_p
    arent"android:id="@+id/imageVie
    w"/>

< /RelativeLayout >
```

#### Code for MainActivity.java:

```
package com.example.exno4;
import android.app.Activity;
import
android.graphics.Bitmap;
import
android.graphics.Canvas;

import
android.graphics.Color;
importandroid.graphics.Pain
t;

import android.graphics.drawable.BitmapDrawable;

import android.os.Bundle;
importandroid.widget.ImageView;
```

```

public class MainActivity extends Activity
{
    @Override
    public void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);

        //Creating a Bitmap
        Bitmap bg = Bitmap.createBitmap(720, 1280, Bitmap.Config.ARGB_8888);

        //Setting the Bitmap as background for the
        ImageView imageView = (ImageView)
        findViewById(R.id.imageView); imageView.setBackgroundDrawable(
        new BitmapDrawable(bg)); //Creating the Canvas Object

        Canvas canvas = new Canvas(bg);

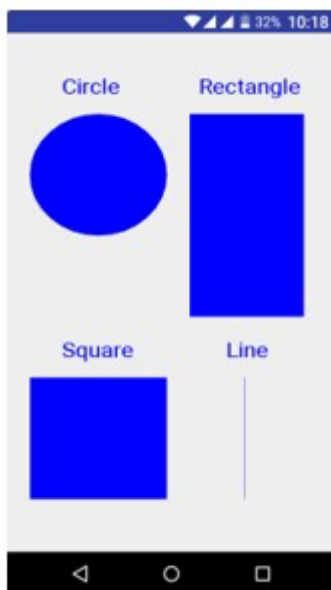
        //Creating the Paint Object and set its color
        Paint paint = new Paint(); paint.setColor(Color.BLUE); paint.set
        TextSize(50);

        //To draw a
        canvas.drawText("Rectangle",
        400, 150, paint); canvas.drawRect(400, 200, 650, 700, paint);

```

```
//Todrawa Circle  
    canvas.drawText("Circle",120,150,paint);  
    canvas.drawCircle(200,350,150,paint);  
//Todrawa Square  
    canvas.drawText("Square",120,800,paint);  
    canvas.drawRect(50,850,350,1150,paint);  
//Todrawa Line  
    canvas.drawText("Line",480,800,paint);  
  
    canvas.drawLine(520,850,520,1150,paint);  
}  
}
```

## Output:



## PART B

(PART B: TO BE COMPLETED BY STUDENTS)

*(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)*

|                              |                             |
|------------------------------|-----------------------------|
| Roll. No. : C-54             | Name: Quazi Md Moinuddin    |
| Class : TE Comps C           | Batch: C-3                  |
| Date of Experiment: 26.03.25 | Date of Submission:01.04.25 |
| Grade:                       |                             |

B.1 Software Code written by student/steps:

### Android\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</RelativeLayout>
```

### MainActivity.java

```
package com.example.shapesapp;

import android.app.Activity;
import android.graphics.Bitmap;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
```

```

import android.graphics.drawable.BitmapDrawable;
import android.os.Bundle;
import android.widget.ImageView;

public class MainActivity extends Activity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Creating a Bitmap
        Bitmap bg = Bitmap.createBitmap(720, 1280, Bitmap.Config.ARGB_8888);

        // Setting the Bitmap as background for the ImageView
        ImageView i = findViewById(R.id.imageView);
        i.setBackground(new BitmapDrawable(getResources(), bg));

        // Creating the Canvas object
        Canvas canvas = new Canvas(bg);

        // Creating the Paint Object and setting its color & text size
        Paint paint = new Paint();
        paint.setTextSize(50);
        paint.setStrokeWidth(10); // For line thickness

        // Drawing Shapes
        paint.setColor(Color.BLUE);
        canvas.drawText("Rectangle", 420, 150, paint);
        canvas.drawRect(400, 200, 650, 700, paint);

        paint.setColor(Color.RED);
        canvas.drawText("Circle", 100, 150, paint);
        canvas.drawCircle(200, 350, 100, paint);

        paint.setColor(Color.MAGENTA);
        canvas.drawText("Square", 400, 750, paint);
        canvas.drawRect(400, 800, 600, 1000, paint); // Square

        paint.setColor(Color.CYAN);
        canvas.drawText("Vertical Line", 50, 900, paint);
        canvas.drawLine(100, 900, 100, 1200, paint); // Vertical Line
    }
}

```

# AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="ShapesApp"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

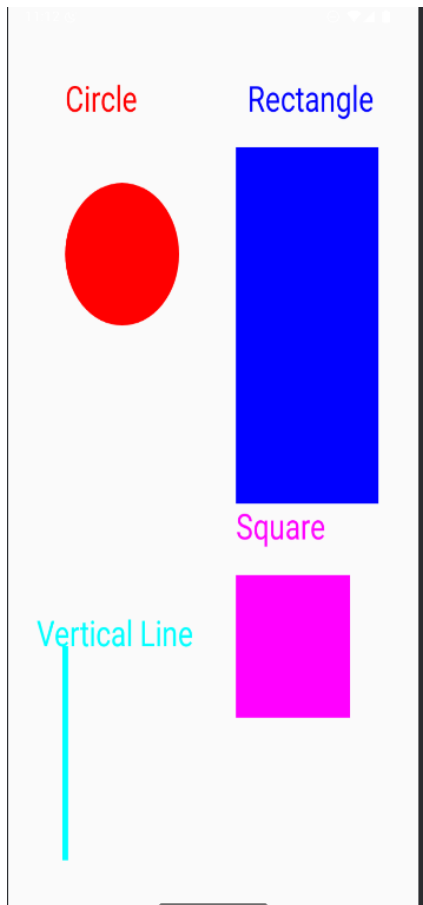
        <activity android:name=".MainActivity"
            android:exported="true"> <!-- Added this line -->

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

    </application>

</manifest>
```

## B.2 Input and Output:



### B.3 Observations and learning:

The application successfully displays a blue rectangle on the screen along with the text "Rectangle" using the **Canvas** and **Paint** classes in Android. The graphical primitives are drawn dynamically on a **Bitmap** and displayed in an **ImageView**.

### B.4 Conclusion:

This experiment demonstrates how to create and manipulate graphical objects in an Android application using the **Canvas API**. It provides a foundation for developing drawing-based applications, such as simple graphic editors or interactive UI elements.

### B.5 Question of Curiosity

#### 1) Explain different steps required to build up this project?

Steps to Build the Shape Drawing Android Project

1. Create a New Project – Open Android Studio, create a new project (**ShapesApp**), and select Java as the language.
2. Modify **activity\_main.xml** – Add an **ImageView** inside a **RelativeLayout** to display the shapes.



3. Update `AndroidManifest.xml` – Ensure `android:exported="true"` is set for Android 12+ compatibility.
4. Write Java Code in `MainActivity.java` –
  - Create a Bitmap and Canvas.
  - Use Paint to draw rectangle, circle, square, horizontal & vertical lines.
  - Set the Bitmap as the background for `ImageView`.
5. Run the Project – Use an emulator or real device to test the app.
6. Enhance & Debug – Fix errors via Logcat, and improve by adding buttons or custom views for dynamic shapes.